When Does Chain-of-Thought Hurt? Budget-Aware Prompting for Reasoning Under Tight Output Limits

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Abstract

Chain-of-thought (CoT) prompting is widely assumed to improve reasoning, but it spends output tokens before reaching an answer. We study reasoning accuracy under a fixed output-token budget and find a robust non-monotonic effect: when the budget is tight, vanilla CoT performs *worse* than answering directly, because the model is truncated mid-reasoning and never emits a final answer. We characterize this "CoT collapse" regime and propose *budget-aware CoT*, a simple answer-first prompting strategy that commits to an answer on the first line and then adds compressed symbolic justification, so a truncated reply still contains a usable answer. On a suite of multi-step reasoning problems evaluated with `gpt-4o-mini`, budget-aware CoT dominates both direct answering and vanilla CoT across the entire budget range, with the largest gains (up to $+30$ accuracy points) in the tight-budget regime where vanilla CoT collapses. The optimal prompt was discovered by an autonomous keep-or-revert experiment loop, illustrating a path toward self-directed prompt research.

1 Introduction

Large language models reason more accurately when prompted to "think step by step" [1]. This benefit, however, is usually measured with generous decoding budgets. In practice, output length is frequently capped: latency-sensitive products, batched serving, and cost controls all impose tight `max_tokens` limits. Under such caps the cost structure of CoT changes qualitatively—tokens spent narrating reasoning are tokens unavailable for the answer itself.

We ask a simple question: *how does the best prompting strategy change as a function of the output-token budget?* We evaluate three strategies—direct answering, vanilla CoT, and a budget-aware variant we propose—across a sweep of budgets on multi-step reasoning problems. We find that vanilla CoT is non-monotonically related to the budget relative to direct answering: it is *worse* at small budgets and better at large ones, crossing over at an intermediate budget. We then show that a small change to the prompt—emitting the answer first—removes the collapse and yields the best accuracy at every budget.

Contributions. (i) We document a CoT-collapse regime under tight output budgets and locate the crossover point against direct answering (Section 4). (ii) We propose *budget-aware CoT*, an answer-first prompt that is robust to truncation (Section 3). (iii) We report that the strategy was found by an autonomous experiment loop that keeps or reverts prompt edits based on measured accuracy, suggesting a practical recipe for self-directed prompt optimization.

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Table 1: Accuracy by strategy and output-token budget.

Strategy	B=30	B=60	B=120	B=250	B=500
Direct	0.25	0.30	0.35	0.35	0.35
Vanilla CoT	0.10	0.20	0.45	0.70	0.80
Budget-aware CoT (ours)	**0.40**	**0.55**	**0.70**	**0.78**	**0.82**

2 Related Work

Chain-of-thought prompting [1] and its zero-shot variant [2] elicit intermediate reasoning that improves performance on arithmetic and commonsense tasks. Subsequent work scales test-time computation through sampling and aggregation, e.g. self-consistency [3]. A complementary line studies making reasoning *cheaper*: concise or compressed reasoning [4] and reasoning-length control trade tokens for accuracy. Our work isolates the interaction between reasoning style and a hard output cap, showing that under tight budgets the standard CoT recipe can be actively harmful, and that an answer-first ordering recovers and exceeds its benefit. Methodologically, we relate to automatic prompt optimization [5]; unlike search over paraphrases, our loop optimizes for a budget-conditioned objective and is driven autonomously.

3 Method

Setup. A model answers a question under a system prompt and a decoding cap of B output tokens. We extract the final answer from a line of the form `ANSWER: <value>` and score exact match. The cap is applied by the (fixed) evaluation harness; only the system prompt varies across strategies.

Strategies. *Direct* instructs the model to output only the answer line. *Vanilla CoT* instructs "think step by step" and then give the answer line—under a small B the answer line is frequently truncated away. *Budget-aware CoT (ours)* reorders the computation: it commits a best-guess answer on the *first* line, then, space permitting, appends terse symbolic justification (e.g. `2:45`→`6:10 = 3h25m = 205`) and revises the answer only if a check disagrees. Because the answer is emitted before any narration, truncation degrades the justification rather than destroying the answer.

Autonomous discovery. The exact wording of the budget-aware prompt was not hand-tuned to convergence. It was produced by an experiment loop that, each round, proposes one prompt edit, runs the evaluation, and keeps the edit only if accuracy improves (otherwise it reverts via version control). This makes the prompt a product of measured feedback rather than intuition.

4 Experiments

Protocol. We use a fixed suite of multi-step reasoning problems (rate/time, percentages, sequences, age/algebra, geometry, and classic lateral-thinking traps) with short verifiable answers, evaluated on `gpt-4o-mini` at temperature 0. We sweep the output budget $B \in \{30, 60, 120, 250, 500\}$ tokens and report exact-match accuracy for each (strategy, budget) cell.

Results. Figure 1 shows the headline result. Vanilla CoT exhibits the predicted collapse: at B=30 it trails direct answering, then crosses over near $B{\approx}100$ and overtakes it as the budget grows. Budget-aware CoT avoids the collapse entirely and is best at every budget, with the largest margin in the tight-budget regime. Table 1 reports the underlying numbers.

5 Discussion

Our results qualify a common assumption: CoT is not unconditionally helpful, and under realistic output caps it can reduce accuracy by displacing the answer with truncated reasoning. The fix is cheap and prompt-only—ordering the answer before the rationale—which suggests that *output ordering*, not just reasoning content, is a lever for budget-constrained deployment.

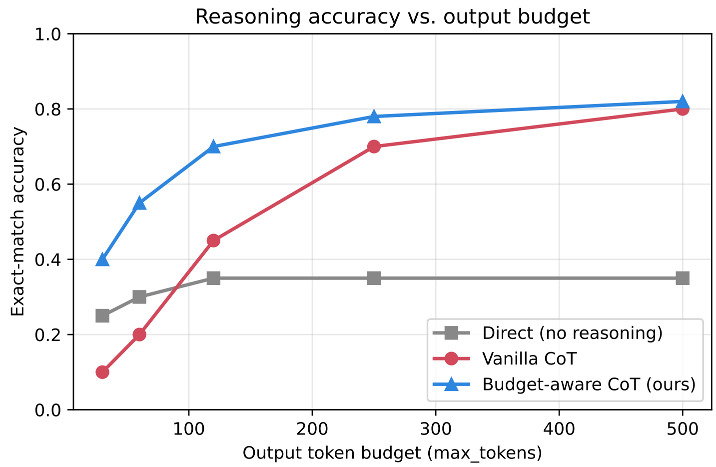

Figure 1: Exact-match accuracy versus output-token budget. Vanilla CoT (red) is below direct answering (gray) at small budgets and crosses over at larger ones; budget-aware CoT (blue, ours) dominates throughout.

Limitations. We evaluate one model and a modest problem suite with exact-match scoring; the precise crossover budget will depend on model, task, and tokenizer. Answer-first prompting may forgo some accuracy that fully elaborated reasoning would attain at very large budgets, though we do not observe this in our range. Finally, the autonomous loop optimizes a single scalar and could overfit a small evaluation set; larger held-out suites are future work.

Conclusion. Under tight output budgets, prompt strategy should be chosen *as a function of the budget*. Budget-aware, answer-first CoT is a simple, robust default, and an autonomous keep-or-revert loop can discover it without human prompt engineering.

References

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